

Student

I keep hearing about **data modeling**, **data warehouses**, **OLTP**, **OLAP**, **star schemas**. My head is spinning. Can you start from zero and build me up to a professional foundation?

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Perfect. This chapter is your **launchpad** for everything that follows in this refcard. You'll learn:

1. What **data modeling** actually is (and isn't).
2. The three **levels of modeling** - conceptual, logical, physical.
3. What a **data warehouse** is and how it differs from a transactional database.
4. **OLTP vs OLAP** - the fundamental split in data workloads.
5. **Normalization** and **why analytics deliberately denormalizes**.
6. **Natural vs surrogate keys** - and why professional data modelers overwhelmingly pick surrogate keys.
7. A **hands-on mini e-commerce model** you can build in any SQL database.

CONCEPT CARD

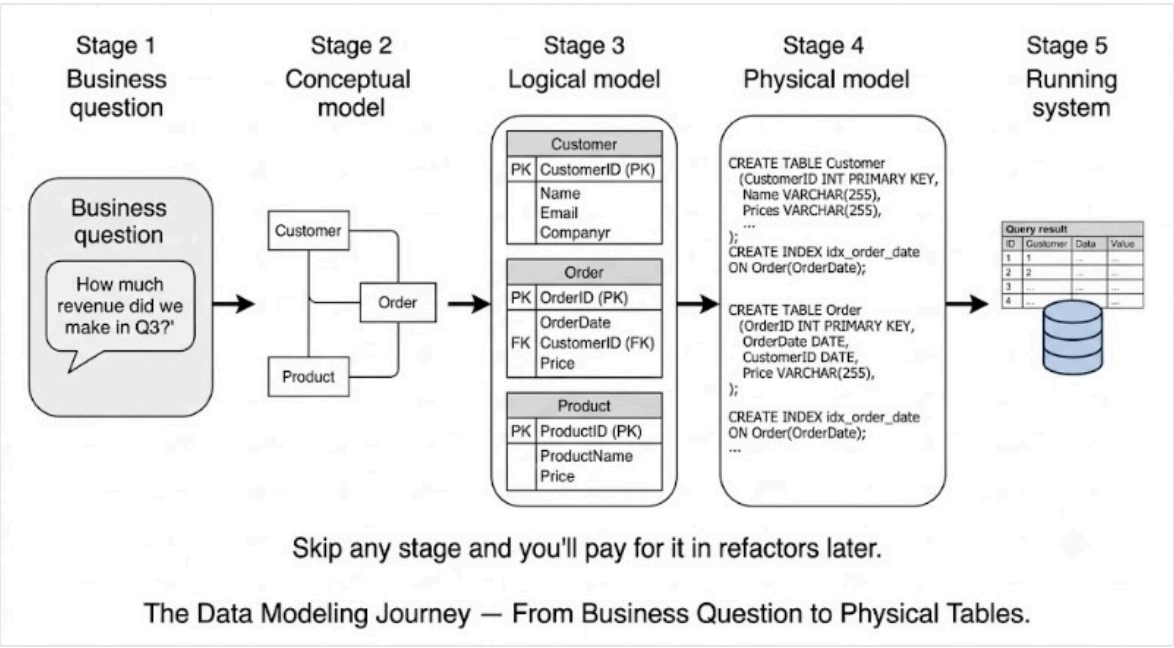
Data modeling is the process of designing how data is **organized**, **stored**, and **related** so it can be queried efficiently and stays consistent over time. A good data model is like a **blueprint before construction**: you spend hours on it because it saves months of pain later.

SUMMARY CARD

By the end of this chapter you'll be able to:

- Distinguish conceptual, logical, and physical models.
- Explain OLTP vs OLAP to any interviewer in 30 seconds.
- Justify denormalization for analytics.
- Design a small warehouse-ready schema with surrogate keys.
- Sketch a real e-commerce model from customer → order → product.

[AI Prompt] Warm-up (paste into any AI chatbot): *"Explain what data modeling is to a beginner. Use an architectural blueprint analogy and give one example of what happens when a system is built without a data model."*



The Data Modeling Journey — From Business Question to Physical Tables

The Three Levels of Data Modeling at a Glance

Level	What It Answers	Who Reads It
Conceptual	What entities and relationships exist?	Business + tech together
Logical	What attributes, keys, and cardinalities?	Data architects
Physical	How is it laid out in this specific DB?	Engineers + DBAs

sql – The end-goal – a small physical model built from a conceptual sketch

```
-- Physical model for a mini e-commerce store.
-- Notice: surrogate keys, explicit data types, indexes.

CREATE TABLE customer (
  customer_key    BIGINT      PRIMARY KEY,
  customer_id     VARCHAR(50) UNIQUE NOT NULL,  -- natural key
  full_name       VARCHAR(200),
  email           VARCHAR(200),
  created_at      TIMESTAMP
);

CREATE TABLE product (
  product_key     BIGINT      PRIMARY KEY,
  sku             VARCHAR(50) UNIQUE NOT NULL,  -- natural key
  product_name    VARCHAR(200),
  unit_price      DECIMAL(12, 2)
```

```
);

CREATE TABLE orders (
  order_key      BIGINT      PRIMARY KEY,
  order_id       VARCHAR(50) UNIQUE NOT NULL,
  customer_key   BIGINT      REFERENCES customer(customer_key),
  order_date     DATE,
  total_amount   DECIMAL(12, 2)
);

CREATE INDEX idx_orders_customer ON orders(customer_key);
CREATE INDEX idx_orders_date     ON orders(order_date);
```

WHAT IS DATA MODELING?

Student

Give me a precise definition of **data modeling**.

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Data modeling is the process of designing how data is **structured, stored, and related** in a system. It's not writing queries, and it's not moving data around (that's ETL). It's the **blueprint** - deciding what tables exist, what columns they have, and how they connect.

CONCEPT CARD

Three things data modeling accomplishes:

1. **Efficiency** - the right model makes queries fast; the wrong model makes them slow (or impossible).
2. **Integrity** - good models enforce rules like *"an order must belong to a customer."*
3. **Communication** - a diagram of a data model is how business and engineering teams agree on what the system means.

Three things data modeling is NOT:

1. **Not writing SQL** - SQL is what you run **on top of** the model.

2. **Not ETL** - pipelines move data into the model; modeling designs the destination.
3. **Not database administration** - DBAs tune what modelers designed.

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What happens without a data model?

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EXAMPLE CARD

Real symptoms of no data model:

1. **Duplicate customers** - same person, three rows, three spellings.
2. **Orders without prices** - nobody thought to make `unit_price` required.
3. **Reports that never balance** - two teams count revenue differently.
4. **Renaming a column breaks 40 dashboards** - nobody knew the dashboards existed.
5. **A schema migration takes 6 months** - the shape of data leaked into every corner of the code.

🔗 ANALOGY

Data modeling is the **architectural blueprint** for a data system. You *could* build a house without a blueprint. But the walls won't be square, the plumbing will leak, and adding a second floor later becomes impossible.

⚠️ WARNING

The most expensive data model is the one you didn't design. Ad-hoc growth almost always produces schemas that cost 10× more to fix than to design correctly upfront.

⚠️ COMMON PITFALL / MISTAKE

Incorrect Way: *"Let's just store everything as JSON blobs and figure out structure later."* Six months in, nobody can query anything.

Correct Way: Spend a few hours modeling the core 5-10 entities up front. Add JSON columns only for genuinely flexible fields.

EXAM TIP CARD

Interview classic: *"What is data modeling?"* Winning answer: *"The process of designing how data is structured, stored, and related - the blueprint that determines whether queries are fast, data stays consistent, and teams share the same meaning of terms like 'customer' and 'order'."*

 EXERCISE

Pick any app you use (Netflix, Amazon, your bank). List the 5-7 core entities it probably has: users, orders, items, payments, addresses, etc. That list is the start of a conceptual model.

[AI Prompt] Paste this into any AI chatbot: *"I'm building a food-delivery app. List the 8 core entities it needs, describe how they relate, and warn me about the top 3 modeling mistakes beginners make in this domain."*

 PRACTICE QUIZ

Which of these is NOT part of data modeling: writing SQL queries, choosing tables and columns, defining relationships between tables, deciding primary keys?

Answer: Writing SQL queries - that's *using* the model, not designing it.

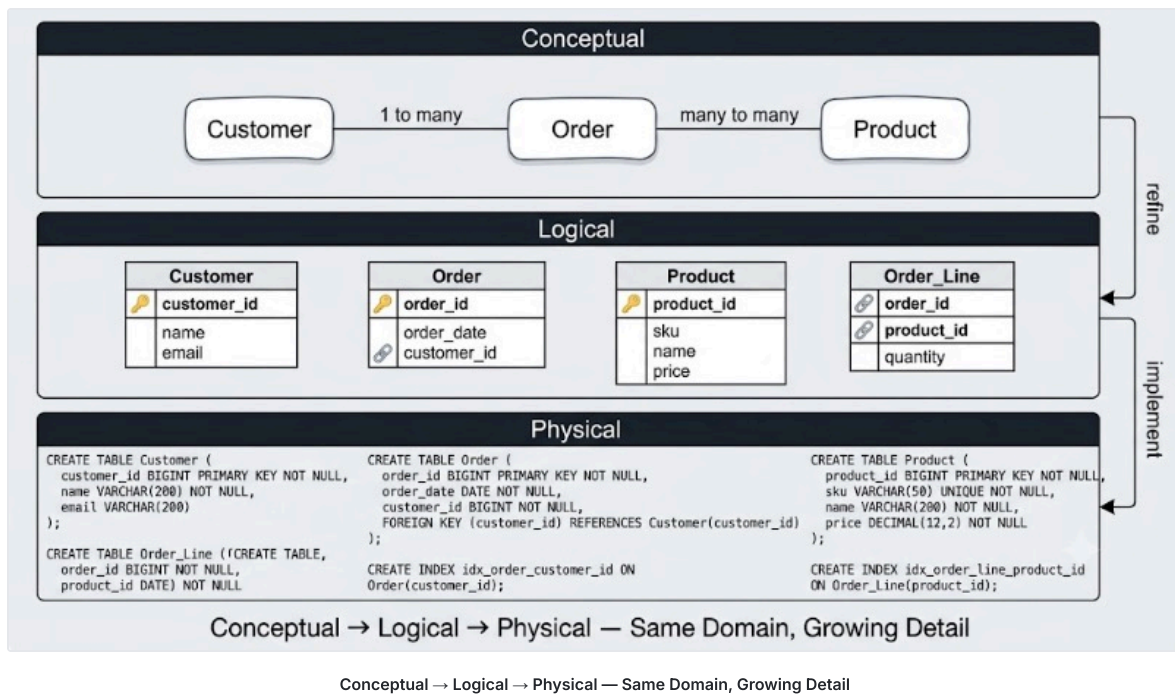
 Reveal Correct Answer

Answer Checklist: Check calculations, verify logic, check edge cases.

THE THREE LEVELS — CONCEPTUAL, LOGICAL, PHYSICAL



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I've read that data models come in **three levels**. What are they?

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Every professional data model is built in three passes, each more detailed than the last:

1. **Conceptual** - *what entities exist and how do they relate?* (Business language.)
2. **Logical** - *what attributes, keys, and cardinalities?* (Platform-independent detail.)
3. **Physical** - *what tables, types, indexes, and constraints in this specific DB?* (Implementation-ready.)

CONCEPT CARD

The progression:

- **Conceptual** is 5-10 boxes on a whiteboard. "A Customer places many Orders. Each Order contains many Products."
- **Logical** adds every attribute, primary keys, foreign keys, and 1-to-many / many-to-many details. Still no SQL.

- **Physical** writes CREATE TABLE with real types, indexes, and constraints. Ready to run.

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Is it always three passes?

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EXAMPLE CARD

The three-pass discipline pays off because each level catches different mistakes:

1. **Conceptual** catches missing entities ("oh, we forgot to model Refund").
2. **Logical** catches missing attributes and unclear relationships ("is a Product in one Category or many?").
3. **Physical** catches performance and integrity issues ("this join needs an index; that column should be NOT NULL").

Skipping a level means catching its mistakes in production.

🔗 ANALOGY

Building a house:

- **Conceptual** = sketch on a napkin ("3 bedrooms, 2 baths, kitchen").
- **Logical** = architect's floor plan (rooms with dimensions, doors, windows).
- **Physical** = engineering blueprints (electrical, plumbing, materials).

Each step is a refinement, not a replacement.

⚠️ WARNING

Don't design the physical model first. It locks you into DB-specific choices before you understand the business. Always: conceptual → logical → physical.

🚫 COMMON PITFALL / MISTAKE

Incorrect Way: Jumping straight to **CREATE TABLE**. You'll ship, then discover a missing entity, then rewrite everything.

Correct Way: 30 minutes on conceptual, 1 hour on logical, then physical. The upfront cost saves weeks later.

EXAM TIP CARD

Interview classic: *"Walk me through the three levels of data modeling."* Winning answer: *"Conceptual = business entities and relationships. Logical = attributes, keys,*

cardinality, still platform-independent. Physical = DB-specific tables, types, indexes."

🔥 EXERCISE

Take a mini domain - say, a library. Sketch the conceptual model on paper: 4-5 boxes for Book, Member, Loan, etc. Then write out the logical attributes for each. Then translate to **CREATE TABLE**. Notice how each level catches a new question.

[AI Prompt] Paste this into any AI chatbot: *"For a movie streaming service (like Netflix), walk me through the conceptual, logical, and physical models for the core Movie/User/Watch domain. Show all three levels with 3-4 entities each."*

WHAT IS A DATA WAREHOUSE?

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What is a **data warehouse**, and how is it different from the database that runs my app?

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A **data warehouse** is a large, centralized store of **integrated data from many sources**, designed and optimized for **analytics and reporting** - not for running an app. Your app's database (the "OLTP" store) is optimized for lots of small transactions; the warehouse is optimized for large queries that summarize months or years of data.

CONCEPT CARD

Core characteristics of a data warehouse:

1. **Integrated** - combines data from many source systems (billing, CRM, product, marketing).
2. **Subject-oriented** - organized by business subjects like Customer, Sales, Product - not by application.
3. **Time-variant** - keeps historical versions of data ("what did our customer count look like last quarter?").
4. **Non-volatile** - data is loaded, not constantly updated in place.
5. **Analytics-optimized** - schemas and storage tuned for reads over huge scans.

Who invented the concept?

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EXAMPLE CARD

Two schools, both still relevant:

1. **Bill Inmon (1990s)** - the "corporate information factory" - build a large enterprise-wide 3NF warehouse first, derive data marts from it.
2. **Ralph Kimball (1996)** - the "dimensional bus" - build subject-focused star schemas (data marts) that share conformed dimensions.
3. Kimball is more common today because it's faster to start, easier to teach, and matches how BI tools query data.

🔗 ANALOGY

If your app database is a **grocery store** (rapid checkout, small transactions), the warehouse is a **library** (organized, historical, designed for research, not for buying eggs).

⚠️ WARNING

Never run analytics queries directly on your production app database. A single 30-minute report can block real user transactions, take down your app, and lose you money.

❗ COMMON PITFALL / MISTAKE

Incorrect Way: Building dashboards on the live app database because "it's easier." Then user requests slow down every time an analyst runs a report.

Correct Way: Load data into a warehouse (or lakehouse) via ETL/ELT. Analysts query there; the app never gets touched.

EXAMPLE CARD

Common data warehouse platforms in 2026:

1. **Snowflake** - cloud-native, separates storage and compute.
2. **Google BigQuery** - serverless, SQL-native, huge scale.
3. **Amazon Redshift** - mature AWS-native warehouse.
4. **Databricks SQL** - warehouse capabilities on lakehouse storage.
5. **Azure Synapse** - Microsoft's warehouse + lakehouse offering.

EXAM TIP CARD

Interview classic: *"What is a data warehouse?"* Winning answer: *"An integrated, subject-oriented, time-variant, non-volatile store optimized for analytics - separated from operational systems so analytics doesn't slow down the app."*

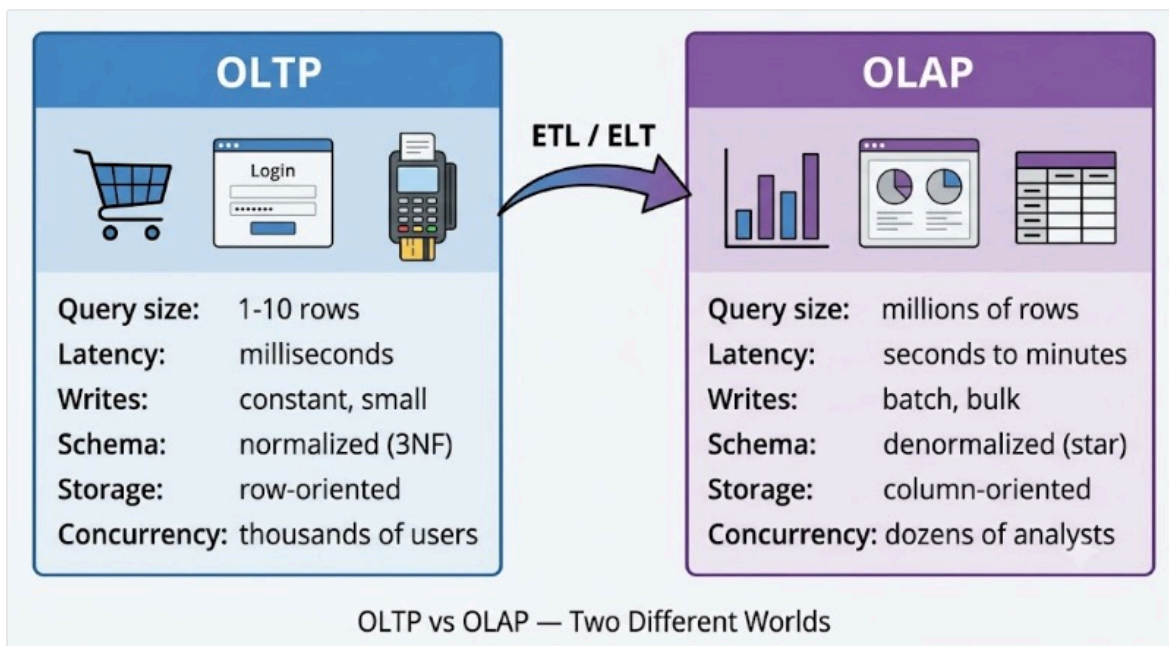
EXERCISE

For any company you know, list 3-4 data source systems that would feed a warehouse (e.g., Stripe for payments, Salesforce for sales, Segment for events, an internal Postgres for product data). That's the raw material for a warehouse.

[AI Prompt] Paste this into any AI chatbot: *"Compare Snowflake, BigQuery, and Redshift as data warehouses in 2026. For each, describe the pricing model, main strengths, and one common gotcha. Format as a 3-column table."*

OLTP VS OLAP

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OLTP vs OLAP — Two Different Worlds

```
sql — OLTP-style query — quick, targeted, small result
```

```
-- OLTP: fetch one user's current shopping cart.
-- Runs in milliseconds. Returns a handful of rows.
```

```
SELECT p.product_name,
```

```

        ci.quantity,
        ci.unit_price
FROM    cart_item ci
JOIN    product p    ON p.product_id = ci.product_id
WHERE   ci.cart_id = 78421
        AND ci.deleted_at IS NULL;

```

sql – OLAP-style query – scans months of data, aggregates by category

```

-- OLAP: revenue by product category for Q3 2026.
-- Scans millions of order lines. Returns a small aggregate.

SELECT p.category,
       SUM(ol.quantity * ol.unit_price) AS revenue,
       COUNT(DISTINCT o.customer_key)  AS unique_buyers
FROM   fact_order_line ol
JOIN   dim_product   p ON p.product_key = ol.product_key
JOIN   fact_order     o ON o.order_key   = ol.order_key
JOIN   dim_date       d ON d.date_key    = o.order_date_key
WHERE  d.year = 2026
      AND d.quarter = 3
GROUP BY p.category
ORDER BY revenue DESC;

```

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I keep seeing **OLTP** and **OLAP**. What are they?

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They describe two fundamentally different **workloads** - the shape of queries a system handles:

- **OLTP (Online Transaction Processing)** - many small, quick, targeted reads/writes. Your bank app updating balances, your e-commerce site placing orders.
- **OLAP (Online Analytical Processing)** - few large, aggregating queries scanning millions of rows. "What was our revenue by product category last quarter?"

CONCEPT CARD

Six axes where they differ:

1. **Query size** - OLTP touches 1-10 rows; OLAP touches millions.
2. **Latency needs** - OLTP demands milliseconds; OLAP tolerates seconds/minutes.
3. **Write pattern** - OLTP: constant small writes; OLAP: batch bulk loads.
4. **Schema style** - OLTP: normalized (3NF); OLAP: denormalized (star schema).
5. **Storage layout** - OLTP: row-oriented; OLAP: column-oriented.
6. **Concurrent users** - OLTP: thousands; OLAP: dozens to hundreds.

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Are OLTP and OLAP different databases?

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EXAMPLE CARD

Usually yes:

1. **OLTP databases** - PostgreSQL, MySQL, MongoDB, Oracle, SQL Server.
2. **OLAP databases** - Snowflake, BigQuery, Redshift, ClickHouse, DuckDB.
3. Data flows from OLTP → warehouse via ETL/ELT (Fivetran, Airbyte, dbt).
4. Some modern engines (like PostgreSQL with columnar extensions, or Databricks) blur the line - HTAP (Hybrid Transactional/Analytical Processing).

🔗 ANALOGY

OLTP is a **fast-food drive-through** - many small orders, each served in seconds.

OLAP is a **restaurant catering a wedding** - few orders, but each serves 200 people and takes hours to prep.

⚠️ WARNING

Running OLAP queries on an OLTP database (or vice versa) is like asking the drive-through to cater the wedding. It might work once. It's not a plan.

🚫 COMMON PITFALL / MISTAKE

Incorrect Way: Storing analytics events in the same table as user profiles. Simple reports lock user updates.

Correct Way: Analytics events go to a warehouse (or lakehouse). The app database stays lean.

EXAM TIP CARD

Interview classic: *"Explain OLTP vs OLAP."* Winning answer: *"OLTP handles many small transactional reads/writes with low latency - normalized, row-store. OLAP handles few large analytical queries over history - denormalized, column-store.*

Different workloads → different databases."

🔥 EXERCISE

For your favorite app, list two OLTP queries (e.g., 'get my current cart', 'add item to cart') and two OLAP queries (e.g., 'top 10 products sold by category last month', 'monthly active users trend'). Notice the shape difference.

[AI Prompt] Paste this into any AI chatbot: *"Give me 3 concrete OLTP query examples and 3 concrete OLAP query examples for an e-commerce site. Show the SQL for each and explain which system would run it and why."*

NORMALIZATION IN 3 MINUTES

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sql – Un-normalized table (WRONG) vs 3NF (RIGHT)

```
-- =====
-- WRONG: un-normalized, redundant, breaks 1NF & 3NF
-- =====

CREATE TABLE orders_bad (
  order_id      INT PRIMARY KEY,
  customer_id   INT,
  customer_name VARCHAR(200),    -- repeated per order (breaks 3NF)
  customer_city VARCHAR(100),    -- repeated per order (breaks 3NF)
  product_ids   VARCHAR(500),    -- 'p1,p2,p3'      (breaks 1NF)
  product_names VARCHAR(500)     -- 'X,Y,Z'        (breaks 1NF)
);

-- =====
-- RIGHT: 3NF – every fact lives in one place
-- =====

CREATE TABLE customer (
  customer_id   INT PRIMARY KEY,
  name          VARCHAR(200),
  city          VARCHAR(100)
);
```

```
CREATE TABLE product (  
  product_id    INT PRIMARY KEY,  
  name          VARCHAR(200)  
);  
  
CREATE TABLE orders (  
  order_id      INT PRIMARY KEY,  
  customer_id   INT REFERENCES customer(customer_id),  
  order_date    DATE  
);  
  
CREATE TABLE order_line (  
  order_id      INT REFERENCES orders(order_id),  
  product_id    INT REFERENCES product(product_id),  
  quantity      INT,  
  PRIMARY KEY (order_id, product_id)  
);
```

Student

Before we talk about denormalization, remind me - what is **normalization**?

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Normalization is the process of organizing tables to **eliminate redundancy and enforce integrity**. It's expressed as a series of "normal forms" - 1NF, 2NF, 3NF, and beyond. For OLTP systems, aiming for **3NF** is the standard.

CONCEPT CARD

The three forms every developer should know:

1. **1NF (First Normal Form)** - each column holds a single value; no repeating groups (no comma-separated lists in a column).
2. **2NF** - 1NF PLUS every non-key column depends on the *whole* primary key (matters for composite keys).
3. **3NF** - 2NF PLUS no transitive dependencies (non-key columns depend *only* on the key, not on other non-key columns).

Student

Give me a quick before/after.

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EXAMPLE CARD

Un-normalized order table (bad):

code — Code Example

```
Order(order_id, customer_id, customer_name, customer_city,  
      product_ids='p1,p2,p3', product_names='X,Y,Z')
```

Problems: comma-separated lists (breaks 1NF), customer info duplicated across every order (breaks 3NF).

3NF version (good):

code — Code Example

```
Customer(customer_id, name, city)  
Product(product_id, name)  
Order(order_id, customer_id)  
Order_Line(order_id, product_id, quantity)
```

Each fact stored once. Customer city change → update one row.

🔗 ANALOGY

Normalization is **decluttering your closet by category**. Each shirt lives in exactly one spot. When you need to update where shirts go, you change one label, not fifty.

⚠️ WARNING

Normalization is a **tool**, not a religion. Beyond 3NF (BCNF, 4NF, 5NF) is rarely worth it in practice. And for analytics, you deliberately *undo* normalization (next subsection).

⚠️ COMMON PITFALL / MISTAKE

Incorrect Way: Storing repeated data "for convenience" in an OLTP table. One year later, updates get inconsistent - some rows have the old city, some the new one.

Correct Way: Normalize to 3NF in OLTP. Any single fact lives in exactly one row.

EXAM TIP CARD

Interview classic: "What is 3NF?" Winning answer: "Every non-key column depends only on the primary key - nothing else. It eliminates transitive dependencies and update anomalies."

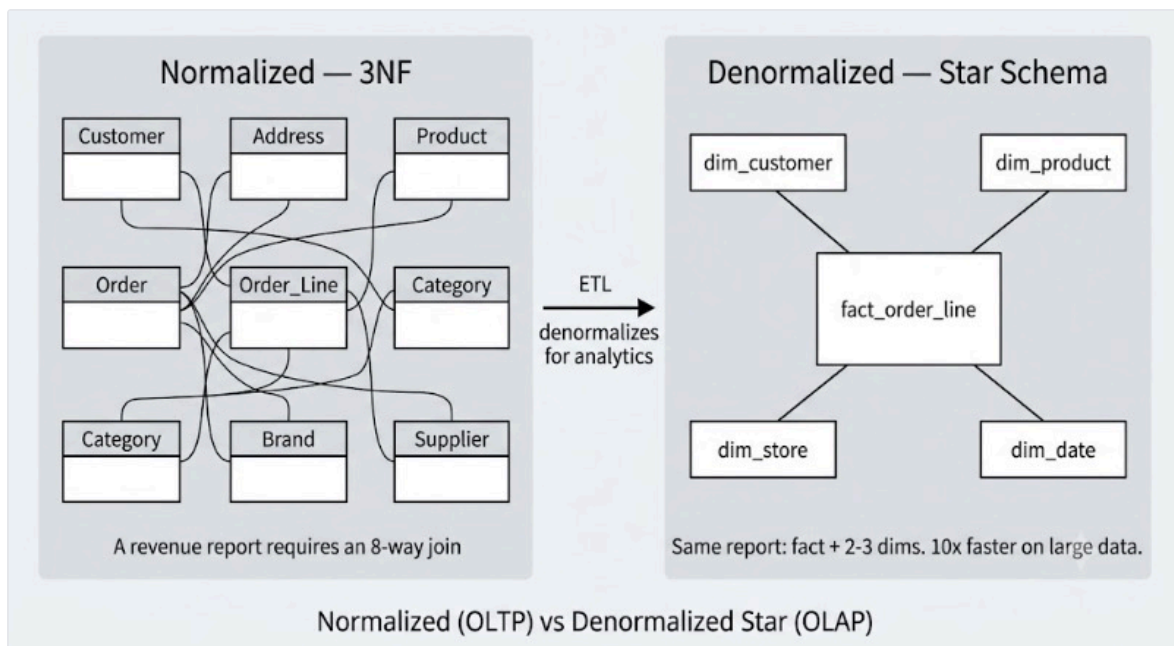
🔥 EXERCISE

Take a spreadsheet you've made recently. Look for repeated data - same customer name in many rows, same address duplicated. That's un-normalized. Sketch what a 3NF version would look like.

[AI Prompt] Paste this into any AI chatbot: "Give me one un-normalized table (like a spreadsheet with repeated customer info) and walk it through 1NF, 2NF, and 3NF step by step. Show the final SQL DDL."

WHY DENORMALIZE FOR ANALYTICS

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Normalized (OLTP) vs Denormalized Star (OLAP)

sql – Normalized (many joins) vs denormalized (few joins) for the same report

```
-- =====
-- NORMALIZED (3NF): revenue by product category – 5-way join
-- =====
SELECT c.name AS category,
```

```

        SUM(ol.quantity * p.unit_price) AS revenue
FROM    order_line ol
JOIN    orders o    ON o.order_id    = ol.order_id
JOIN    product p   ON p.product_id  = ol.product_id
JOIN    category c  ON c.category_id = p.category_id
JOIN    customer u  ON u.customer_id = o.customer_id
WHERE   o.order_date ≥ '2026-07-01'
GROUP  BY c.name;

-- =====
-- DENORMALIZED (star): same answer, simpler and faster
-- category name is already IN dim_product, no extra join.
-- =====

SELECT p.category,
       SUM(ol.line_revenue) AS revenue
FROM   fact_order_line ol
JOIN   dim_product p ON p.product_key = ol.product_key
JOIN   dim_date     d ON d.date_key   = ol.order_date_key
WHERE  d.year_month ≥ '2026-07'
GROUP BY p.category;

```

Student

If normalization is so good, why do warehouses **denormalize**?

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Because OLTP and OLAP optimize for **opposite goals**. Normalization is great for **frequent writes** and **integrity**.

But for analytics you have **infrequent bulk writes** and **frequent huge reads** - the trade-off flips. Denormalization trades **storage** and **write cost** for **query speed** and **simplicity**.

CONCEPT CARD

What you gain by denormalizing for analytics:

1. **Fewer joins** - a report scans one big fact table + a few dimensions instead of 8 tables.
2. **Query simplicity** - analysts don't need to know 20 tables to write a report.
3. **Query speed** - fewer joins = faster on huge datasets.

4. **Predictable performance** - the star schema (Chapter 4) enables consistent BI performance.
5. **Historical accuracy** - you can freeze a customer's city as it was at order time.

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What do I lose?

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⚠ WARNING

Trade-offs of denormalization:

1. **Redundant storage** - customer name copied to every fact row.
2. **Update complexity** - updating a customer means updating many rows (usually handled by SCD Type 2 - Chapter 3).
3. **Risk of inconsistency** - if two rows say different things, which is right?
4. In OLAP, we accept these because **writes are batch** and **integrity is enforced by the ETL**, not by the schema.

EXAMPLE CARD

The two big denormalization patterns for analytics:

1. **Star schema** - one central fact table with many dimensions (Chapter 4).
2. **One Big Table (OBT)** - everything for a subject flattened into one wide table. Simpler for BI tools; harder to maintain.

💡 ANALOGY

A normalized OLTP model is a **card catalog** - every book referenced once, cross-referenced. A denormalized OLAP model is a **printed encyclopedia** - the same fact appears in many articles because you're reading, not editing.

⚠ COMMON PITFALL / MISTAKE

Incorrect Way: Applying 3NF to your warehouse because "that's what you learned in databases class." Reports become 10-way joins that take minutes.

Correct Way: Denormalize deliberately using star schema. Your reports get 2× to 100× faster.

EXAM TIP CARD

Winning interview line: *"Normalization optimizes for write efficiency and integrity - perfect for OLTP. Denormalization optimizes for read speed and query simplicity - the right choice for OLAP. Use the right tool for the workload."*

🔥 EXERCISE

Sketch a normalized OLTP schema for a bookstore (customer, order, orderline, book, author, category). Then sketch a denormalized fact table for "order lines" with dimension attributes baked in (title, categoryname, author_name). Notice how much shorter reports become.

[AI Prompt] Paste this into any AI chatbot: *"Show me a normalized (3NF) e-commerce schema and its denormalized star-schema equivalent for analytics. Write one query against each showing the difference in complexity."*

NATURAL KEYS — WHAT THEY ARE

Student

What's a **natural key**?

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A **natural key** is a real-world identifier that already exists for the entity - like an **email address**, a **social security number**, a **book's ISBN**, or a **product SKU**. It's not invented by the database; it comes from the business or the outside world.

CONCEPT CARD

Common natural keys:

1. **Email address** - for a user.
2. **SSN / PAN / Aadhaar** - for a person (with privacy caveats).
3. **ISBN** - for a book.
4. **SKU** - for a product.
5. **Order number** - issued by the business.
6. **VIN** - for a car.

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Why not always use natural keys?

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EXAMPLE CARD

Real problems with natural keys:

1. **They can change** - email addresses get updated; phone numbers reassigned; SKUs revised.
2. **They can be reused** - a discontinued SKU may return for a new product years later.
3. **They're often PII** - using SSN or email as a PK leaks sensitive data into logs, indexes, and every foreign key.
4. **They're big and slow** - a 200-character email is a worse PK than a small integer.
5. **Composite naturals hurt** - sometimes you need 3 columns to uniquely identify a row; every FK then becomes 3 columns.

💡 ANALOGY

Using a natural key as a primary key is like using someone's **street address** as their forever ID. They move → your whole database now points to a wrong address.

⚠️ WARNING

Some natural keys are legally sensitive. Using SSN or Aadhaar as a primary key can create real compliance problems - those fields end up in every audit log and backup.

⚠️ COMMON PITFALL / MISTAKE

Incorrect Way: Using `email` as the primary key of a `customer` table. The user changes emails; you now have to update every foreign key that referenced them.

Correct Way: Use a surrogate integer PK; keep the email as a `UNIQUE` column.

EXAM TIP CARD

Interview classic: *"What's a natural key?"* Winning answer: *"An identifier that already exists in the real world - like an email, SKU, or VIN. Useful for lookups, but rarely the best primary key because it can change, get reused, or leak PII."*

🔥 EXERCISE

For any table you've designed, ask: *"What's the natural key?"* - the thing a human would use to identify a row. Then ask: *"Could it ever change?"* If yes, the natural key shouldn't be the PK.

[AI Prompt] Paste this into any AI chatbot: *"List 5 domains (e.g., healthcare, publishing, retail, transport, education) and for each identify a common natural key, its main weakness, and why a surrogate PK is preferred in practice."*

SURROGATE KEYS — WHY WE PREFER THEM

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sql – Dimension table with BOTH surrogate PK and natural key

```
CREATE TABLE dim_customer (  
    customer_key      BIGINT          PRIMARY KEY,  -- surrogate PK  
    customer_id       VARCHAR(50)    NOT NULL,      -- natural key from CRM  
  
    -- Descriptive attributes  
    full_name         VARCHAR(200),  
    email             VARCHAR(200),  
    city              VARCHAR(100),  
    country           VARCHAR(80),  
  
    -- SCD Type 2 columns (Chapter 3)  
    effective_from    DATE            NOT NULL,  
    effective_to      DATE,  
    is_current        BOOLEAN         NOT NULL DEFAULT TRUE  
);  
  
-- Natural key is UNIQUE per active row, not across ALL rows.  
-- (Multiple historical rows can share the same customer_id.)  
CREATE UNIQUE INDEX ux_dim_customer_active  
    ON dim_customer(customer_id)  
    WHERE is_current = TRUE;  
  
-- Speed up joins from fact tables on the surrogate key  
CREATE INDEX idx_dim_customer_key  
    ON dim_customer(customer_key);
```

Student

What's a **surrogate key**?

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A **surrogate key** is a **system-generated identifier** with no business meaning - usually an auto-increment integer or a UUID. It exists purely to give each row a stable, single-column, small, unique identifier that never changes.

CONCEPT CARD

Why professional modelers pick surrogate keys almost every time:

1. **Stable** - never changes, even if the natural key does.
2. **Small** - typically a **BIGINT** ; faster joins and smaller indexes than a **VARCHAR(200)** .
3. **Single column** - no composite key headaches in foreign keys.
4. **No PII** - no legal or ethical baggage.
5. **Decouples from source systems** - if your CRM restarts its ID sequence, your warehouse is unaffected.
6. **Enables SCD Type 2** (Chapter 3) - you can have two rows for the same natural entity, one "active" and one "historical."

Student

Auto-increment integer or UUID?

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EXAMPLE CARD

Trade-offs:

1. **Auto-increment BIGINT** - small, fast, sequential; hurts sharding and distributed systems.
2. **UUID (v4)** - globally unique; safer for merging data from many sources; larger (16 bytes vs 8).
3. **UUID v7 (time-ordered)** - best of both - unique across systems, still index-friendly.
4. In warehouses, **auto-increment BIGINT** is by far the most common. Use UUIDs only when merging cross-source data.

🔗 ANALOGY

Natural keys are like **nicknames** - meaningful to humans, but they can change. Surrogate keys are like **serial numbers** - meaningless to humans, but perfectly stable.

⚠️ WARNING

Never expose surrogate keys in URLs or APIs unless it's fine to reveal your row-count and growth rate. Use a separate opaque identifier (like a UUID) for external references.

⚠️ COMMON PITFALL / MISTAKE

Incorrect Way: Building a dimension table with only the natural key (`product SKU`) as the PK. When the SKU changes, history breaks.

Correct Way: `product_key BIGINT PRIMARY KEY` (surrogate) + `sku VARCHAR(50) UNIQUE NOT NULL` (natural key kept as an attribute).

EXAM TIP CARD

Interview classic: *"Why do dimensional models use surrogate keys?"* Winning answer: *"They give each row a stable, small, non-PII identifier that never changes. That enables historical tracking (SCD Type 2), decouples us from source systems, and speeds up joins on large fact tables."*

🔥 EXERCISE

Take a table you've designed. Add a `_key BIGINT` surrogate PK. Keep the old natural key as a `UNIQUE` column. Notice: no downstream code that references the surrogate key ever has to change, even if the natural key does.

[AI Prompt] Paste this into any AI chatbot: *"Write DDL for a dimension table representing a Product with both a surrogate key (product_key) and the natural key (SKU). Add columns for name, category, price, and effective date range. Include the UNIQUE and NOT NULL constraints."*

CHOOSING BETWEEN NATURAL AND SURROGATE KEYS

Student

When should I actually use each?

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For **dimension tables in a warehouse**, always use a **surrogate PK** and keep the natural key as an attribute. For **OLTP tables**, use whichever is genuinely stable - most modern apps prefer surrogates here too.

The important thing is **consistent naming** and **explicit intent**.

CONCEPT CARD

The professional pattern:

1. `**<entity>\_key**` = surrogate integer/BIGINT PK. Used in every join.
2. `**<entity>\_id**` = natural key from the source system. Kept as an attribute; often UNIQUE.
3. Foreign keys always reference the surrogate **_key** , never the natural **_id** .
4. Analytics tools and BI dashboards join on **_key** .
5. Support engineers look up rows by **_id** (the human-readable one).

Student

When could a natural key still be the PK?

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EXAMPLE CARD

Rare cases where natural keys work as PKs:

1. **Static reference data** - country codes (ISO 3166), currency codes (ISO 4217) - these truly never change.
2. **Junction tables** - the composite (*orderkey*, *productkey*) is fine as PK for the line table.
3. **Extremely simple systems** - small internal apps with 3 tables.
4. Even here, adding a surrogate rarely hurts and future-proofs you.

🔗 ANALOGY

Natural and surrogate keys serve different jobs. The natural key is your **passport name** - how the world identifies you. The surrogate key is your **social security number in the database's system** - meaningless outside, but perfectly stable for internal use.

⚠️ WARNING

Never mix conventions. Half your tables using natural PKs and half using surrogates is a nightmare to query. Pick a rule (usually: surrogate everywhere) and stick to it.

⚠ COMMON PITFALL / MISTAKE

Incorrect Way: Naming both the surrogate and the natural key `id` - nobody can tell them apart in queries.

Correct Way: Clear naming - `_key` for surrogate, `_id` for natural.

EXAM TIP CARD

Interview classic: *"Should every table have a surrogate key?"* Winning answer: *"For warehouse dimensions and most OLTP tables, yes - it decouples from source systems, keeps FKs small, and enables historical tracking. Exceptions are pure junction tables and static reference data."*

🔥 EXERCISE

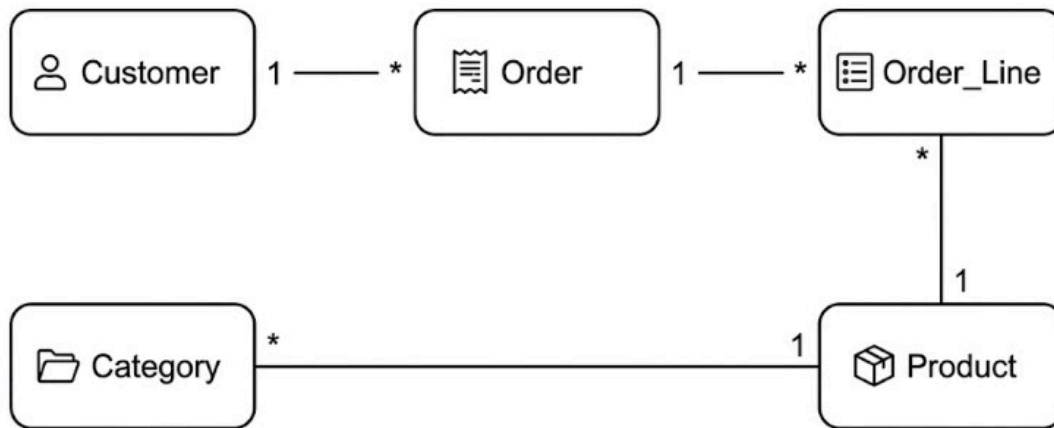
Look at any schema you've built. For each PK, ask: *"Natural or surrogate?"* If any natural key COULD change, plan the migration to surrogate. Do it before the natural key actually changes.

[AI Prompt] Paste this into any AI chatbot: *"Design a small warehouse schema (customer, product, order, order_line) using the surrogate key + natural key pattern with proper naming conventions. Show the DDL and explain each choice."*

HANDS-ON — MODEL A MINI E-COMMERCE STORE



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* = many, 1 = one, and the FK direction follows the cardinality.

Mini E-Commerce Conceptual Model.

Mini E-Commerce Conceptual Model

sql – Complete mini e-commerce OLTP schema – copy, paste, run

```

-- =====
-- Mini e-commerce store – OLTP schema (3NF-ish)
-- Surrogate PKs, natural keys kept as UNIQUE columns.
-- =====

-- Category: reference data for product classification
CREATE TABLE category (
    category_key    BIGINT        PRIMARY KEY,
    category_code   VARCHAR(20)   UNIQUE NOT NULL,    -- e.g.,
    'ELECTRONICS'
    category_name   VARCHAR(100)  NOT NULL
);

-- Customer: end user
CREATE TABLE customer (
    customer_key    BIGINT        PRIMARY KEY,
    customer_id     VARCHAR(50)   UNIQUE NOT NULL,    -- natural key from
    signup
    full_name       VARCHAR(200)  NOT NULL,
    email           VARCHAR(200)  UNIQUE,
    city            VARCHAR(100),
    country_code    CHAR(2),
    created_ts      TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP
);
  
```

```

-- Product: what we sell
CREATE TABLE product (
    product_key    BIGINT        PRIMARY KEY,
    sku            VARCHAR(50)    UNIQUE NOT NULL,    -- natural key
    product_name    VARCHAR(200) NOT NULL,
    category_key    BIGINT        NOT NULL
                    REFERENCES category(category_key),
    unit_price      DECIMAL(12, 2) NOT NULL CHECK (unit_price ≥ 0),
    is_active       BOOLEAN       NOT NULL DEFAULT TRUE
);

-- Order header
CREATE TABLE orders (
    order_key       BIGINT        PRIMARY KEY,
    order_id        VARCHAR(50)    UNIQUE NOT NULL,    -- natural key from
OMS
    customer_key    BIGINT        NOT NULL
                    REFERENCES customer(customer_key),
    order_ts        TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
    order_status    VARCHAR(20)    NOT NULL DEFAULT 'PLACED'
);

-- Order line: many products per order
CREATE TABLE order_line (
    order_line_key  BIGINT        PRIMARY KEY,
    order_key       BIGINT        NOT NULL
                    REFERENCES orders(order_key),
    product_key     BIGINT        NOT NULL
                    REFERENCES product(product_key),
    quantity        INT           NOT NULL CHECK (quantity > 0),
    unit_price      DECIMAL(12, 2) NOT NULL,    -- snapshot price at
order time
    line_amount     DECIMAL(12, 2) NOT NULL
);

-- Indexes for common query paths
CREATE INDEX idx_orders_customer ON orders(customer_key);
CREATE INDEX idx_orders_ts       ON orders(order_ts);
CREATE INDEX idx_line_order      ON order_line(order_key);
CREATE INDEX idx_line_product    ON order_line(product_key);

-- =====
-- Verify the model with a small query
-- =====

-- Customer names and total spent per order
SELECT c.full_name,

```

```
        o.order_id,  
        o.order_ts,  
        SUM(l.line_amount) AS order_total  
FROM    orders      o  
JOIN    customer    c ON c.customer_key = o.customer_key  
JOIN    order_line  l ON l.order_key     = o.order_key  
GROUP   BY c.full_name, o.order_id, o.order_ts  
ORDER   BY o.order_ts DESC;
```

Student

Can you show me a **complete example** that ties everything from this chapter together?

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Here's a **mini e-commerce domain** built end-to-end. We'll go conceptual → logical → physical, and end with SQL you can run in any modern database (PostgreSQL, Snowflake, BigQuery - all work).

CONCEPT CARD

What this hands-on demonstrates:

1. Conceptual sketch (5 entities + relationships).
2. Logical model (attributes, keys, cardinality).
3. Physical DDL - with surrogate keys, natural keys, indexes, and constraints.
4. Naming conventions applied consistently.
5. A tiny sample query at the end to prove it works.

Student

Is this OLTP or OLAP?

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EXAMPLE CARD

This first version is **OLTP** (3NF-ish). That's the starting point of most real projects. Chapter 4 will show you how to convert this into a **star schema** for analytics.

🔗 ANALOGY

Think of this as your **base recipe**. Every future chapter - Kimball facts and dimensions, SCD, star vs snowflake, lakehouse patterns, feature stores - will remix this same domain to teach you different techniques.

⚠️ WARNING

For a real production model, you'd add: audit columns (*createdat*, *updatedat*, *created_by*), soft-delete flags, more constraints (CHECK, ENUMs), and often partitioning on large tables. This example keeps it minimal for clarity.

🔥 EXERCISE

Copy the DDL below into any SQL environment. Run it. Insert 2 customers, 3 products, and 2 orders. Write a query that returns customer name + order date + total. That's a full end-to-end model exercise in 30 minutes.

[AI Prompt] Paste this into any AI chatbot: *"Review my e-commerce schema for the conceptual → logical → physical progression. Point out any missing entities, weak keys, or missing constraints. Suggest what would change if I moved to a star schema for analytics."*

CHAPTER REVIEW & SUMMARY

Student

Can you explain what this chapter is about?

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CHAPTER SUMMARY

This chapter laid the foundation for every advanced technique in the rest of this refcard - Kimball dimensional modeling, SCD, star vs snowflake, lakehouse patterns, and feature stores all build on these concepts.

- **WHAT IS DATA MODELING:** The process of designing how data is structured, stored, and related - the blueprint that makes queries fast, keeps data consistent, and gives teams a shared vocabulary for the business.

- THREE LEVELS: Conceptual (entities and relationships), logical (attributes, keys, cardinality), physical (DDL, types, indexes) - always progress in that order.
- WHAT IS A DATA WAREHOUSE: An integrated, subject-oriented, time-variant, non-volatile store optimized for analytics - separated from operational systems so reports never slow down the app.
- OLTP VS OLAP: OLTP handles many small transactional reads/writes; OLAP handles few large analytical queries - different query shapes, different schemas, different databases.
- NORMALIZATION IN 3 MINUTES: 1NF (atomic values), 2NF (depend on the whole PK), 3NF (no transitive dependencies) - the OLTP standard.
- WHY DENORMALIZE FOR ANALYTICS: Trade storage and write cost for query speed and simplicity - the fundamental move behind star schemas.
- NATURAL KEYS: Real-world identifiers like email, SKU, or ISBN - useful for lookups but risky as PKs because they change, get reused, or leak PII.
- SURROGATE KEYS: System-generated identifiers (typically BIGINT) - stable, small, PII-free, and the key to SCD Type 2 later in the refcard.
- CHOOSING BETWEEN KEYS: Surrogate PK + natural key as UNIQUE attribute is the professional default; use consistent naming (*key for surrogate*, *id* for natural).
- HANDS-ON MINI E-COMMERCE: A complete 5-entity OLTP schema (category, customer, product, orders, order_line) with surrogate keys, natural keys, constraints, and indexes - the base we'll remix in every future chapter.

Student

Can you break down the next concept?

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You now have the foundation every data engineer, analyst, and architect shares. In the next chapter we get to the heart of the refcard: **Kimball - Facts, Dimensions, Grain, and Conformed Dimensions**. That's where the real magic of analytical modeling starts.